MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MICROPROCESSOR, KEY CODE MJA

This amendment forms a part of Military Specification MIL-M-28787/306A(EC), dated 11 August 1981, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

FIGURE 1, CPU block: Delete "INTREQ" and substitute "~~INT~~REQ".

FIGURE 1, CPU block: Delete "JAQ" and substitute "IAQ".

PAGES 4, 5, 6, and 7

* TABLE I: Delete and substitute new table I as printed on pages 3, 4, 5, and 6 of this amendment.

PAGES 28 AND 29

FIGURE 12: Delete and substitute new figure 12 as printed on pages 7 and 8 of this amendment.

PAGE 32

TABLE VIII, Test condition G3, Max column: Delete "175" and substitute "195".

PAGE 33

FIGURE 14: Delete and substitute new figure 14 as printed on page 9 of this amendment.

PAGE 41

FIGURE 16: Delete and substitute new figure 16 as printed on page 10 of this amendment.

PAGE 48

FIGURE 19: Delete and substitute new figure 19 as printed on page 11 of this amendment.

PAGE 55

TABLE XIII, NOTE 4, second line: Delete "V03" and substitute "M02".

PAGE 56

FIGURE 23: Delete and substitute new figure 23 as printed on page 12 of this amendment.



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PAGE 59

* TABLE XIV: Delete and substitute new table XIV as printed on page 13 of this amendment.

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* TABLE XV: Delete and substitute new tables XV-A and XV-B as printed on pages 14 and 15 of this amendment.

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TABLE XVIII, Test condition R31, Parameter column: Delete "PI to IC1" and substitute "P1 to IC1".

PAGE 108

* TABLE XX: Delete and substitute new table XX as printed on page 16 of this amendment.

PAGE 109

* TABLE XXI: Delete and substitute new table XXI as printed on page 17 of this amendment.

PAGE 123

TABLE XXV: Delete and substitute new table XXV as printed on page 18 of this amendment.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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(Project 5963-N488)



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TABLE I. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25°C ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0°C ±5°C | | 60°C ±5°C | | -55°C ±5°C | | 85°C ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| A | Isolation | | | | | | | | | | | | | |
| | Initial | II | S | 10 | ∞ | 10 | ∞ | 10 | ∞ | 10 | ∞ | 10 | ∞ | MΩ |
| | End-of-life | II | S | 1 | ∞ | 1 | ∞ | 1 | ∞ | 1 | ∞ | 1 | ∞ | MΩ |
| B1-B4 | Continuity | III | X | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | Ω |
| C | Power supply stress check | IV | S | ← Must pass all remaining tests → | | | | | | | | | | |
| D1-D16 | Input breakdown current | V | S | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 1.0 | mA |
| D17 | Input breakdown current | V | S | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | 0.0 | 0.1 | mA |
| D18 | Input breakdown current | V | S | 0.0 | 0.2 | 0.0 | 0.2 | 0.0 | 0.2 | 0.0 | 0.2 | 0.0 | 0.2 | mA |
| E1, E3-E5 | Output short circuit current | VI | X | -20 | -55 | -20 | -55 | -20 | -55 | -20 | -55 | -20 | -55 | mA |
| E2,E6,E7 | Output short circuit current | VI | X | -20 | -100 | -20 | -100 | -20 | -100 | -20 | -100 | -20 | -100 | mA |
| E8-E47 | Output short circuit current | VI | X | -9.1 | -12.4 | -9.1 | -12.4 | -9.1 | -12.4 | -9.1 | -12.4 | -9.1 | -12.4 | mA |
| F1-F40 | Input clamp diode voltage | VII | X | -0.2 | -1.5 | -0.2 | -1.5 | -0.2 | -1.5 | -0.2 | -1.5 | -0.2 | -1.5 | V |
| G1 | Injector supply voltage | VIII | X | 0.71 | 1.32 | 0.71 | 1.32 | 0.71 | 1.32 | 0.71 | 1.32 | 0.71 | 1.32 | V |
| G2 | Injector supply voltage | VIII | X | 0.74 | 1.37 | 0.74 | 1.37 | 0.74 | 1.37 | 0.74 | 1.37 | 0.74 | 1.37 | V |
| G3 | $V_{CC}$ power supply current | VIII | X | 20 | 195 | 20 | 195 | 20 | 195 | 20 | 195 | 20 | 195 | mA |
| H1-H6,H8-H14,H16, H18 | Input current, high level | IX | X | 0 | 80 | 0 | 80 | 0 | 80 | 0 | 80 | 0 | 80 | µA |
| H7,H15,H17 | Input current, high level | IX | X | 0 | 40 | 0 | 40 | 0 | 40 | 0 | 40 | 0 | 40 | µA |
| H19 | Input current, high level | IX | X | -2.6 | -3.4 | -2.6 | -3.4 | -2.6 | -3.4 | -2.6 | -3.4 | -2.6 | -3.4 | mA |
| H20-H23,H25 | Input current, high level | IX | X | 0 | 300 | 0 | 300 | 0 | 300 | 0 | 300 | 0 | 300 | µA |
| H24 | Input current, high level | IX | X | 0 | 1000 | 0 | 1000 | 0 | 1000 | 0 | 1000 | 0 | 1000 | µA |
| J1-J6,J8-J14, J16 | Input current, low level | IX | X | -0.2 | -3.50 | -0.2 | -3.50 | -0.2 | -3.50 | -0.2 | -3.50 | -0.2 | -3.50 | mA |
| J7,J15,J18 | Input current, low level | IX | X | -0.1 | -1.75 | -0.1 | -1.75 | -0.1 | -1.75 | -0.1 | -1.75 | -0.1 | -1.75 | mA |
| J17 | Input current, low level | IX | X | -0.1 | -0.88 | -0.1 | -0.88 | -0.1 | -0.88 | -0.1 | -0.88 | -0.1 | -0.88 | mA |
| J19 | Input current, low level | IX | X | -5.0 | -10.8 | -5.0 | -10.8 | -5.0 | -10.8 | -5.0 | -10.8 | -5.0 | -10.8 | mA |
| J20-J23,J25 | Input current, low level | IX | X | 0 | -10 | 0 | -10 | 0 | -10 | 0 | -10 | 0 | -10 | µA |
| J24 | Input current, low level | IX | X | 0 | -20 | 0 | -20 | 0 | -20 | 0 | -20 | 0 | -20 | µA |

See footnotes at end of table.





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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25°C ±5°C 4/ | | Class I 4/- 0°C ±5°C | | Class I 4/- 60°C ±5°C | | Class II 4/ -55°C ±5°C | | Class II 4/ 85°C ±5°C | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| K1-K24 | Output voltage, high level | X | X | 5.22 | 5.55 | 5.22 | 5.55 | 5.22 | 5.55 | 5.22 | 5.55 | 5.22 | 5.55 | V |
| K25-K40 | Output voltage, high level | X | X | 5.05 | 5.55 | 5.05 | 5.55 | 5.05 | 5.55 | 5.05 | 5.55 | 5.05 | 5.55 | V |
| K41-K47 | Output voltage, high level | X | X | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | 2.4 | 4.5 | V |
| L1-L47 | Output voltage, low level | XI | X | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | 0.0 | 0.4 | V |
| M1-M28 | Static functional test, priority interrupt and clock shaping | XII | X | ← See tables XII and XIII → | | | | | | | | | | |
| N1-N4 | Dynamic functional test, minimum and maximum injector current | XIV | X | ← See tables XIV and XV and 6.3 → | | | | | | | | | | |
| P1-P40 | Transition time ($t_{THL}$), $C_L$ = 150 pF | XVI | Q | 5 | 40 | 5 | 40 | 5 | 45 | 5 | 40 | 5 | 47 | ns |
| P41,P43-P45 | Transition time ($t_{THL}$), $C_L$ = 50 pF | XVI | Q | 4 | 50 | 4 | 50 | 4 | 55 | 4 | 50 | 4 | 57 | ns |
| P42,P46,P47 | Transition time ($t_{THL}$), $C_L$ = 50 pF | XVI | Q | 4 | 20 | 4 | 20 | 4 | 25 | 4 | 20 | 4 | 27 | ns |
| Q1,Q3-Q5 | Transition time ($t_{TLH}$), $C_L$ = 50 pF | XVII | Q | 10 | 80 | 10 | 80 | 10 | 90 | 10 | 80 | 10 | 95 | ns |
| Q2,Q6,Q7 | Transition time ($t_{TLH}$), $C_L$ = 50 pF | XVII | Q | 3 | 45 | 3 | 45 | 3 | 55 | 3 | 45 | 3 | 60 | ns |
| R1 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVIII | Q | 10 | 58 | 10 | 60 | 10 | 60 | 10 | 60 | 10 | 61 | ns |
| R2 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVIII | Q | 10 | 43 | 10 | 45 | 10 | 45 | 10 | 45 | 10 | 46 | ns |
| R3-R18,R28, R42,R60 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVIII | Q | 10 | 61 | 10 | 63 | 10 | 63 | 10 | 63 | 10 | 64 | ns |
| R19,R34-R37,R47, R49,R52, R53,R64, R67,R70, R72 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVIII | Q | 10 | 56 | 10 | 58 | 10 | 58 | 10 | 58 | 10 | 59 | ns |
| R20-R27 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVIII | Q | 10 | 51 | 10 | 53 | 10 | 53 | 10 | 53 | 10 | 54 | ns |

See footnotes at end of table.





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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25°C ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0°C ±5°C | | 60°C ±5°C | | -55°C ±5°C | | 85°C ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| R29-R33, R38-R41, R43-R45, R50,R51, R54,R55, R58,R59, R61,R62, R65,R68, R71,R73, R75,R77, R79 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVIII | Q | 10 | 46 | 10 | 48 | 10 | 48 | 10 | 48 | 10 | 49 | ns |
| R46,R48, R56,R57, R63,R66, R69,R74, R76,R78 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVIII | Q | 10 | 50 | 10 | 52 | 10 | 52 | 10 | 52 | 10 | 53 | ns |
| R80-R83 | Propagation delay time ($t_{PLH}$), $C_L$ = 50 pF | XVIII | Q | 5 | 29 | 5 | 31 | 5 | 31 | 5 | 31 | 5 | 32 | ns |
| S1,S28,S42, S60 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIX | Q | 10 | 68 | 10 | 71 | 10 | 71 | 10 | 71 | 10 | 72 | ns |
| S2,S29,S43, S61 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIX | Q | 10 | 38 | 10 | 41 | 10 | 41 | 10 | 41 | 10 | 42 | ns |
| S3-S18 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIX | Q | 10 | 48 | 10 | 51 | 10 | 51 | 10 | 51 | 10 | 52 | ns |
| S19 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIX | Q | 10 | 41 | 10 | 44 | 10 | 44 | 10 | 44 | 10 | 45 | ns |
| S20-S27 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIX | Q | 10 | 36 | 10 | 39 | 10 | 39 | 10 | 39 | 10 | 40 | ns |
| S30-S33, S38-S41, S44,S45, S50,S51, S54,S55, S58,S59, S62,S65, S68,S71, S73,S75, S77,S79 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIX | Q | 10 | 37 | 10 | 40 | 10 | 40 | 10 | 40 | 10 | 41 | ns |

See footnotes at end of table.



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TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/-3/ | 25°C ±5°C 4/ | | Class I 4/ | | | | Class II 4/ | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 0°C ±5°C | | 60°C ±5°C | | -55°C ±5°C | | 85°C ±5°C | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| S34-S37, S47,S49, S52,S53, S64,S67, S70,S72 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIX | Q | 10 | 63 | 10 | 66 | 10 | 66 | 10 | 66 | 10 | 67 | ns |
| S46,S48, S56,S57, S63,S66, S69,S74, S76,S78 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIX | Q | 10 | 42 | 10 | 45 | 10 | 45 | 10 | 45 | 10 | 46 | ns |
| S80-S83 | Propagation delay time ($t_{PHL}$), $C_L$ = 50 pF | XIX | Q | 5 | 44 | 5 | 47 | 5 | 47 | 5 | 47 | 5 | 48 | ns |
| T1,T2 | Dynamic functional test and propagation delay time check | XX | Q | ← See tables XX and XXI and 6.3 → | | | | | | | | | | |
| U1,U3 | Maximum frequency check, clock shaping (PRP2) | XXII | Q | 201 | 207 | 201 | 207 | 201 | 207 | 201 | 207 | 201 | 207 | ns |
| U2,U4 | Maximum frequency check, clock shaping (PWA2) | XXII | Q | 52 | 69 | 52 | 69 | 52 | 69 | 52 | 69 | 52 | 69 | ns |
| V1-V31 | Output crosstalk, low level | XXIII | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| W1-W5 | Output crosstalk, high level | XXIII | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| X | Power supply transient amplitude | XXIV | Q | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | 0 | 400 | mV |
| Y1 | Power supply capacitance, $V_{CC}$ | XXV | S | 3.22 | 5.98 | 3.22 | 5.98 | 3.22 | 5.98 | 3.22 | 5.98 | 3.22 | 5.98 | µF |
| Y2 | Power supply capacitance, INJ | XXV | S | 200 | 1500 | 200 | 1500 | 200 | 1500 | 200 | 1500 | 200 | 1500 | pF |
| Z1-Z16 | Input capacitance | XXV | Q | 2 | 20 | 2 | 20 | 2 | 20 | 2 | 20 | 2 | 20 | pF |
| Z17-Z41 | Input cacpacitance | XXV | Q | 2 | 40 | 2 | 40 | 2 | 40 | 2 | 40 | 2 | 40 | pF |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.



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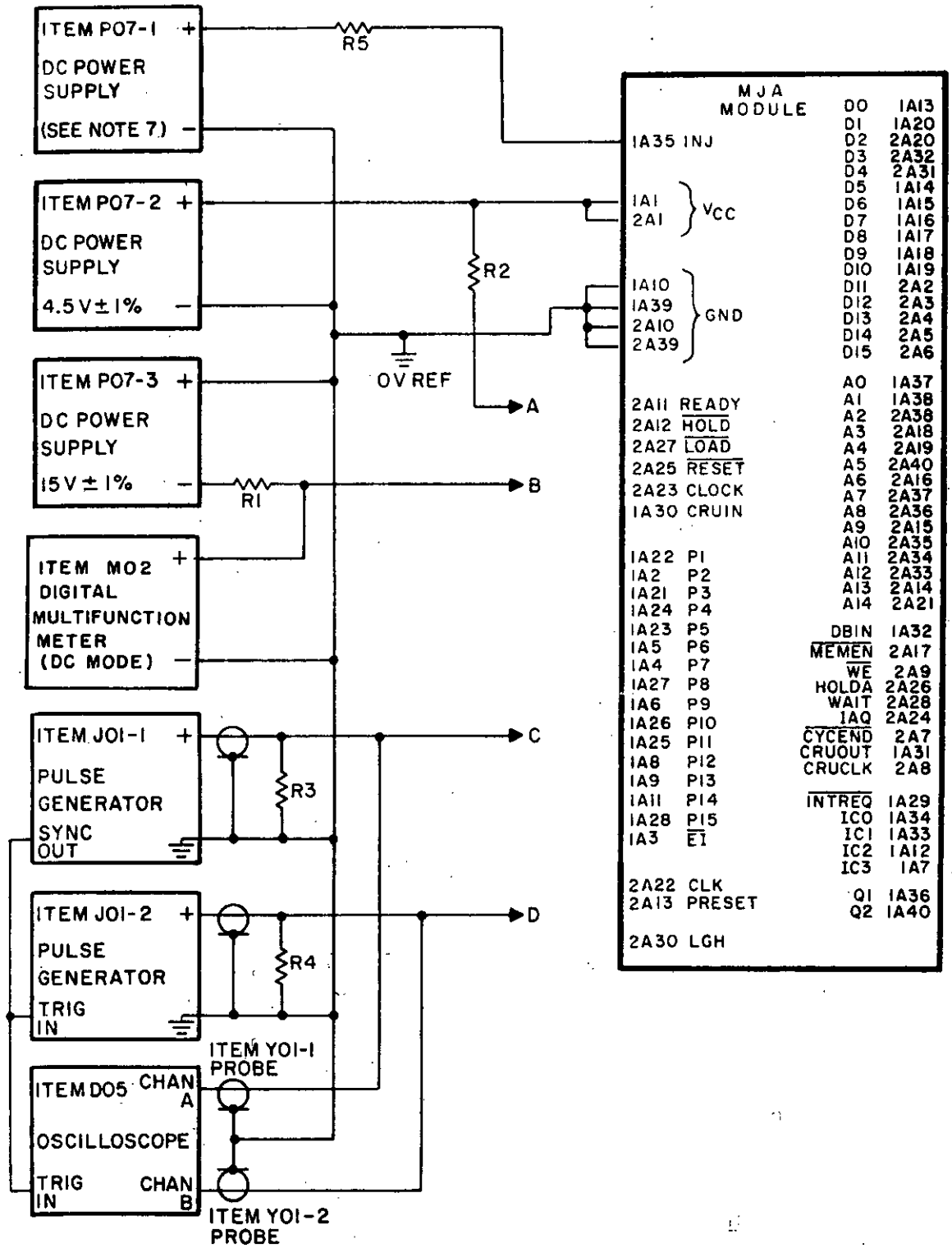



FIGURE 12. Input clamp diode voltage test circuit for table VII.



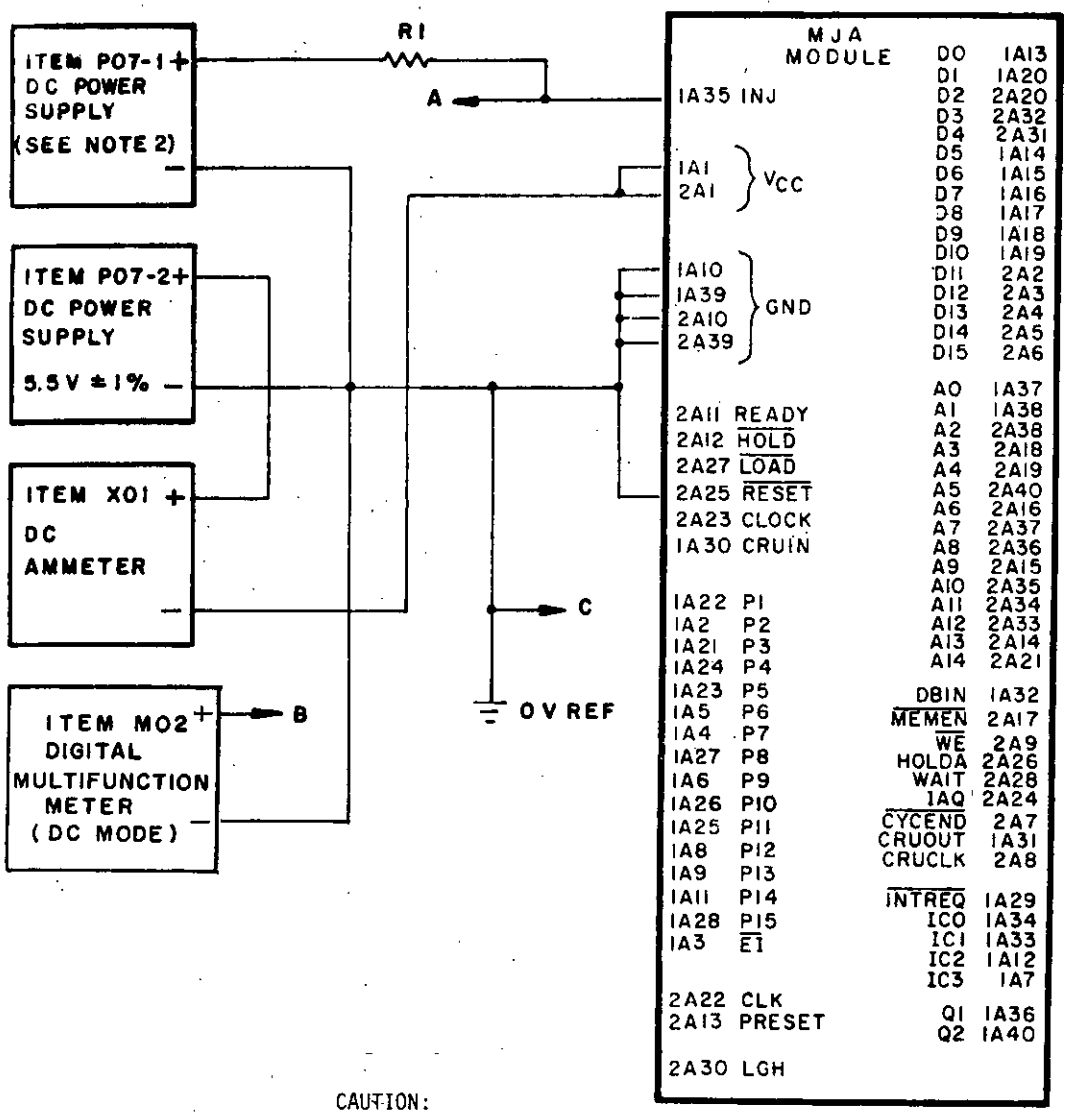
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NOTES:
1. R1 = 1.13 kΩ ±1%, 1/4 W for test conditions F1-F16 and F19-F40.
2. R1 = 750Ω ±1%, 1/4 W for test conditions F17 and F18.
3. R2 = 1.0 kΩ ±5%, 1/4 W.
4. R3 = R4 = 51Ω ±5%, 1/2 W.
5. R5 = 4Ω ±1%, 2 W.
6. For test conditions F25-F40 use the CPU preconditioning flow chart on figure 13 to set the CPU DATA lines into the input mode.
7. Adjust item P07-1 voltage to limit pin 1A35 current to 450 mA ±1%.

FIGURE 12. Input clamp diode voltage test circuit for table VII - Continued.



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CAUTION:

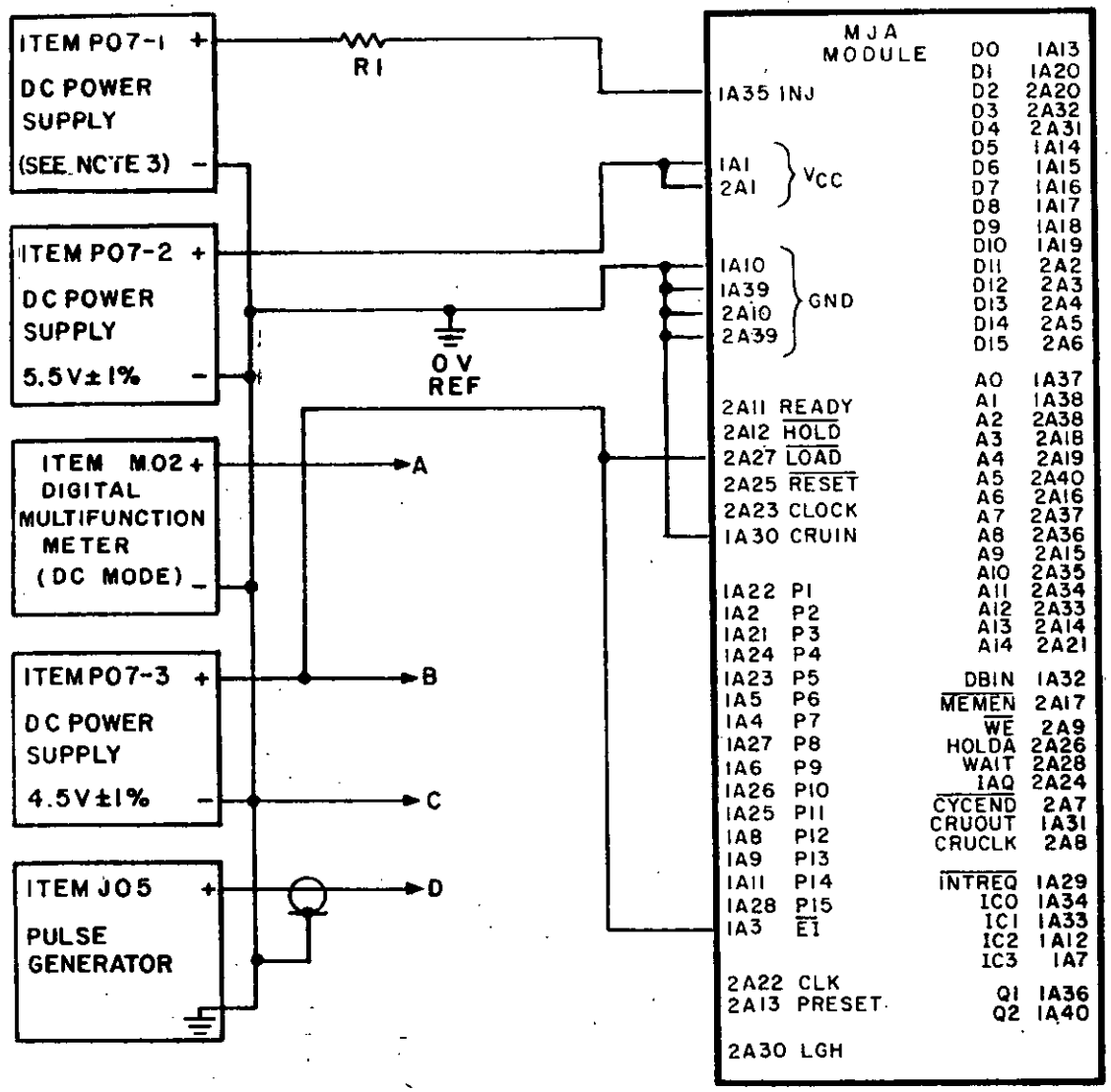
Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 4Ω ±1%, 2 W.
2. Adjust item P07-1 voltage to limit pin 1A35 current to 450 mA ±1% for G1 and 550 mA ±1% for G2. Set to 500 mA for G3.

FIGURE 14. Injector supply voltage and power supply current test circuit for table VIII.



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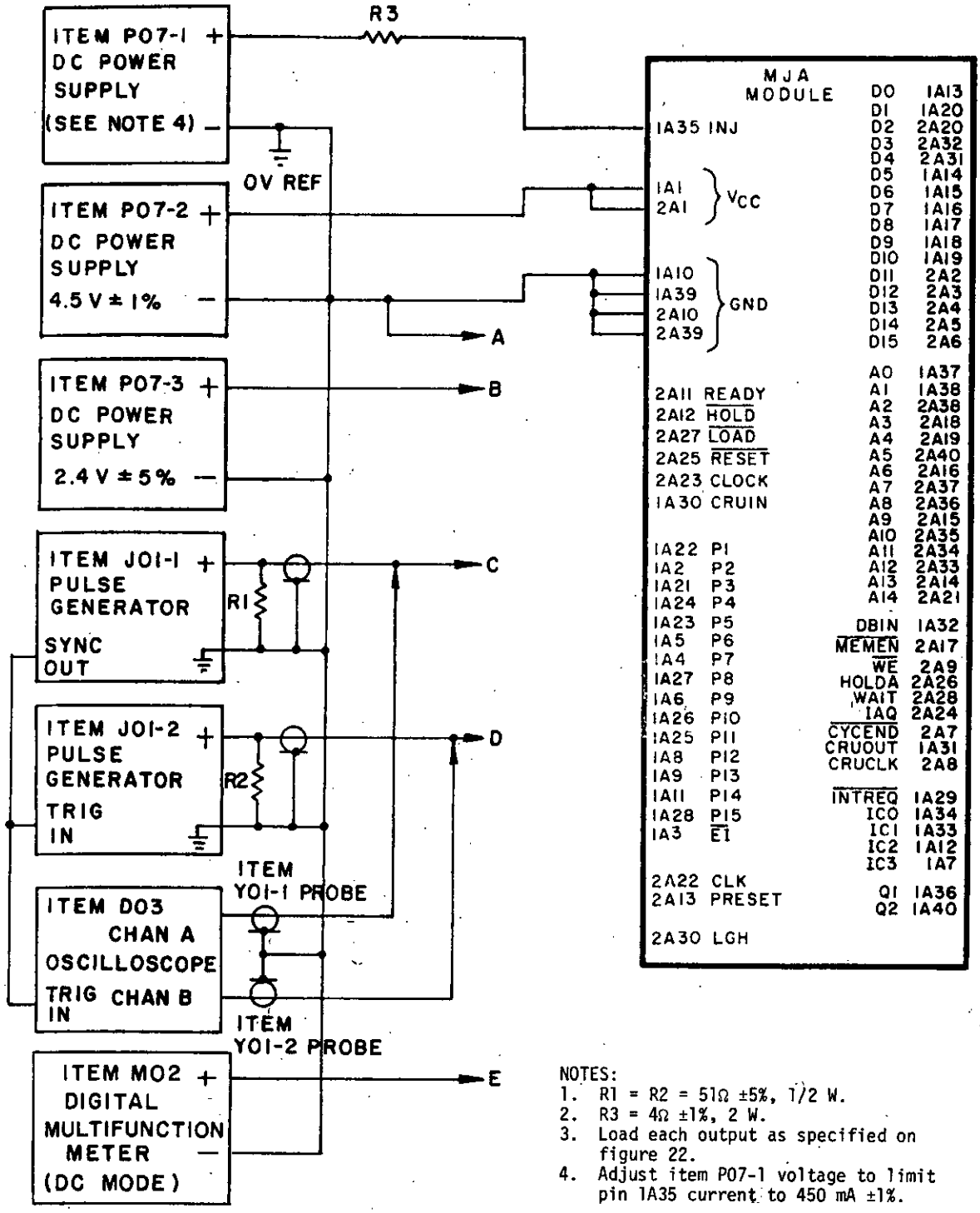



NOTES:
1. R1 = 4Ω ±1%, 2 W.
2. Use the preconditioning pattern of figures 17 and 18.
3. Adjust item P07-1 voltage to limit pin 1A35 current to 550 mA ±1%.

FIGURE 16. Output voltage (high-level) test circuit for test conditions K1-K40 in table X.



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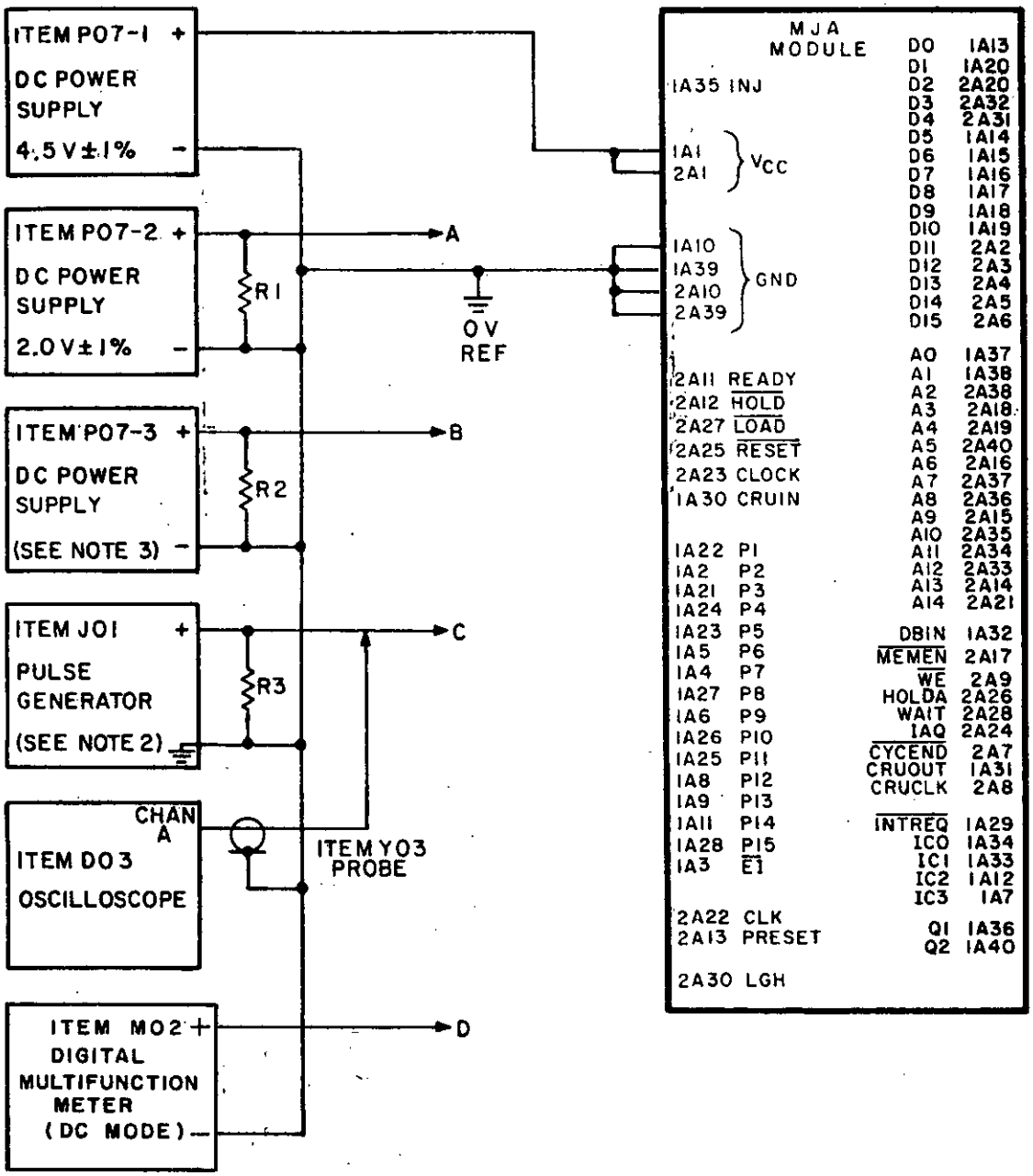



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 4Ω ±1%, 2 W.
3. Load each output as specified on figure 22.
4. Adjust item P07-1 voltage to limit pin 1A35 current to 450 mA ±1%.

FIGURE 19. Output voltage (high-level and low-level) test circuit for test conditions K41-K47 in table X and for test conditions L1-L40 in table XI.



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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Use the flow chart on figure 25 to obtain the required clock pulse numbers and input setups for test conditions M18-M28 in table XII.
3. Item PO7-3 voltage:
   V = 0.8 V ±1% for test conditions M1-M17.
   V = 0.7 V ±1% for test conditions M18-M28.

FIGURE 23. Static functional (priority interrupt and clock shaping) test circuit for table XII.



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TABLE XIV. Dynamic functional (minimum and maximum injector current) test. 1/, 2/

| Test condition | Parameter | Input table | Waveform figure | Test figure |
|---|---|---|---|---|
| N1 | Dynamic functional test, minimum injector current | XV-A | 27 | 26 |
| N2 | Dynamic functional test, maximum injector current | XV-A | 27 | 26 |
| N3 | Dynamic functional test, minimum injector current | XV-B | 27 | 26 |
| N4 | Dynamic functional test, maximum injector current | XV-B | 27 | 26 |

1/ Apply the vectors (see 6.3) for each test condition using the input setup and hold times, power supply voltage and injector current as specified in tables XV-A and XV-B and figure 26. Verify the output levels are $V_{OL} \leq 0.8$ V and $V_{OH} \geq 2.0$ V at a point in time not less than the maximum propagation delay time shown in table XXI for test conditions N1 and N2.

2/ Verify the output levels are $V_{OL} \leq 0.9$ V and $V_{OH} \geq 2.0$ V at a point in time not less than the maximum propagation delay time shown in table XXI for test conditions N3 and N4.





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TABLE XV-A. CPU input conditions for table XIV.

| Parameter | Pin(s) | Input conditions | Unit | Figure |
|---|---|---|---|---|
| INJ | 1A35 | 1/,2/ | mA | 26,27 |
| $V_{CC}$ | 1A1,2A1 | 4.5 ±1% | V | |
| PRP | 2A23 | 1000 ±1% | ns | |
| PWA | 2A23 | 333 ±1% | | |
| $t_{TLH1}$, $t_{TLH2}$, $t_{THL1}$, $t_{THL2}$ | All CPU inputs | 15 ±5 | | |
| $t_S$ | 1A13,1A14,1A15,1A16,1A17,1A18,1A19, 2A2,2A3,2A4,2A5,2A6,2A20,2A31,2A32 | 287 ±10 | | |
| $t_S$ | 2A11 | 292 ±10 | | |
| $t_S$ | 2A12 | 332 ±10 | | |
| $t_S$ | 1A30 | 272 ±10 | | |
| $t_S$ | 1A29 | 247 ±10 | | |
| $t_S$ | 1A34,1A33,1A12,1A7 | 247 ±10 | | |
| $t_H$ | 1A13,1A14,1A15,1A16,1A17,1A18,1A19, 2A2,2A3,2A4,2A5,2A6,2A20,2A31,2A32 | 145 ±10 | | |
| $t_H$ | 2A11 | 170 ±10 | | |
| $t_H$ | 2A12 | 160 ±10 | | |
| $t_H$ | 1A30 | 180 ±10 | | |
| $t_H$ | 1A29 | 160 ±10 | | |
| $t_H$ | 1A34,1A33,1A12,1A7 | 160 ±10 | | |
| $V_{IH}$ | All | 2.0 +0.1, -0.0 | V | |
| $V_{IL}$ | All | 0.6 +0.0, -0.1 | V | |

1/ INJ = 450 mA ±1% for test conditions N1, T1.
2/ INJ = 550 mA ±1% for test conditions N2, T2.





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TABLE XV-B. CPU input conditions for tables XIV and XX.

| Parameter | Pin(s) | Input conditions | Unit | Figure |
|---|---|---|---|---|
| INJ | 1A35 | 1/, 2/ | mA | 26,27 |
| $V_{CC}$ | 1A1,2A1 | 4.5 ±1% | V | |
| PRP | 2A23 | 333 ±1% | ns | |
| PWA | 2A23 | 111 ±1% | | |
| $t_{TLH1}$, $t_{TLH2}$, $t_{THL1}$, $t_{THL2}$ | All CPU inputs | 15 ±5 | | |
| $t_S$ | 1A13,1A14,1A15,1A16,1A17,1A18,1A19, 2A2,2A3,2A4,2A5,2A6,2A20,2A31,2A32 | 65 +2, -0 | | |
| $t_S$ | 2A11 | 70 +2, -0 | | |
| $t_S$ | 2A12 | 110 +2, -0 | | |
| $t_S$ | 1A30 | 50 +2, -0 | | |
| $t_S$ | 1A29 | 25 +2, -0 | | |
| $t_S$ | 1A34,1A33,1A12,1A7 | 25 +2, -0 | | |
| $t_H$ | 1A13,1A14,1A15,1A16,1A17,1A18,1A19, 2A2,2A3,2A4,2A5,2A6,2A20,2A31,2A32 | 40 +2, -0 | | |
| $t_H$ | 2A11 | 50 +2, -0 | | |
| $t_H$ | 2A12 | 25 +2, -0 | | |
| $t_H$ | 1A30 | 40 +2, -0 | | |
| $t_H$ | 1A29 | 55 +2, -0 | | |
| $t_H$ | 1A34,1A33,1A12,1A7 | 55 +2, -0 | | |
| $V_{IH}$ | All | 3.0 +0.1, -0.0 | V | |
| $V_{IL}$ | All | 0.0 +0.0, -0.1 | V | |

1/ INJ = 450 mA ±1% for test conditions N3, T1.
2/ INJ = 550 mA ±1% for test conditions N4, T2.





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TABLE XX. Dynamic functional (minimum and maximum injector current) test and propagation delay time ($t_{PLH}$ and $t_{PHL}$) check. 1/

| Test condition | Test description | Input table | Output table | Waveform | Test figure |
|---|---|---|---|---|---|
| T1 | Dynamic functional test, minimum injector current and propagation delay time check | XV-B | XXI | 51 | 26 |
| T2 | Dynamic functional test, maximum injector current and propagation delay time check | XV-B | XXI | 51 | 26 |

1/ Apply the vectors (see 6.3) with the input parameters of table XV-B and checking the output parameters of table XXI.





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TABLE XXI. CPU output conditions for table XX. 1/, 2/

| Parameter | Pin(s) | Output conditions | | Unit | Test figure |
|---|---|---|---|---|---|
| | | Min | Max | | |
| A0-A14 Propagation delay time | 1A37,1A38,2A14,2A15,2A16,2A17,2A18,2A19,2A21, 2A33,2A34,2A35,2A36,2A37,2A38,2A40 | --- | 140 | ns | 26,51 |
| D0-D15 | 1A13,1A14,1A15,1A16,1A17,1A18,1A19,1A20,2A2,2A3, 2A4,2A5,2A6,2A20,2A31,2A32 | --- | 155 | | |
| DBIN | 1A32 | --- | 160 | | |
| $\overline{\text{MEMEN}}$ | 2A17 | --- | 140 | | |
| $\overline{\text{WE}}$ | 2A28 | --- | 185 | | |
| HOLDA | 2A26 | --- | 290 | | |
| WAIT | 2A28 | --- | 130 | | |
| IAQ | 2A24 | --- | 130 | | |
| $\overline{\text{CYCEND}}$ | 2A7 | --- | 145 | | |
| CRUOUT | 1A31 | --- | 185 | | |
| CRUCLK | 2A8 | --- | 155 | | |
| Output switch voltage | | 1.5 | 1.5 | V | |

1/ For test conditions N1 and N2, the outputs shall be checked with the compare window set at 900 ns and continuing for 10 ns.

2/ For test conditions N3, N4, T1, and T2 check the propagation delay times by starting output comparison at the maximum propagation delay time limits and continuing for 10 ns.



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TABLE XXV. Power supply and input capacitance test - Continued.

| Test condition | Parameter | Connect A to pin | Connect B to pin | Initial limits 25°C ±5°C ambient Min | Initial limits 25°C ±5°C ambient Max | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| Z17 | CLK Input capacitance | Open | 2A22 | 2 | 40 | pF | 59 |
| Z18 | PRESET | | 2A13 | | | | |
| Z19 | LGH | | 2A30 | | | | |
| Z20 | READY | | 2A11 | | | | |
| Z21 | $\overline{\text{HOLD}}$ | | 2A12 | | | | |
| Z22 | $\overline{\text{LOAD}}$ | | 2A27 | | | | |
| Z23 | $\overline{\text{RESET}}$ | | 2A25 | | | | |
| Z24 | CLOCK | | 2A23 | | | | |
| Z25 | CRUIN | | 1A30 | | | | |
| Z26 | D0 | | 1A13 | | | | |
| Z27 | D1 | | 1A20 | | | | |
| Z28 | D2 | | 2A20 | | | | |
| Z29 | D3 | | 2A32 | | | | |
| Z30 | D4 | | 2A31 | | | | |
| Z31 | D5 | | 1A14 | | | | |
| Z32 | D6 | | 1A15 | | | | |
| Z33 | D7 | | 1A16 | | | | |
| Z34 | D8 | | 1A17 | | | | |
| Z35 | D9 | | 1A18 | | | | |
| Z36 | D10 | | 1A19 | | | | |
| Z37 | D11 | | 2A2 | | | | |
| Z38 | D12 | | 2A3 | | | | |
| Z39 | D13 | | 2A4 | | | | |
| Z40 | D14 | | 2A5 | | | | |
| Z41 | D15 | | 2A6 | | | | |





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